Dynamic Semantic Field Theory (DSFT)

A Temporal Framework for Semantic Force Dynamics in Dialogue Systems

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Abstract

We introduce Dynamic Semantic Field Theory (DSFT), a temporal computational framework for modeling dialogue semantics as a system of four interacting forces rather than as static, turn-level classifications. Traditional natural language processing classifiers — including BERT, LSTM, and rule-based keyword systems — treat each utterance in isolation, discarding the temporal dynamics that characterize real human discourse. DSFT proposes that every dialogue generates a tension field composed of four operational semantic forces: Analytical Pressure (F_A), Exploratory Expansion (F_E), Affective Resonance (F_R), and Persuasive Drift (F_P). These forces evolve continuously under equations incorporating inertia, inter-force coupling, momentum, and hysteresis resistance.

The DSFT-TD V2.1 implementation achieves four key results under controlled benchmark conditions: (1) correct four-force dominance classification across all synthetic dialogue types, (2) early transition detection with a latency of 7 turns before the dominance shift occurs, (3) a false alarm rate of 3.3% in stable technical dialogues, and (4) sustained stability over 40+ conversational turns without force collapse. A configurable observer layer introduces four measurement modes — Passive, Active, Reflexive, and Meta — enabling researcher-controlled variation in how the field is sampled. An external falsification protocol (EVALUATION_PROTOCOL v1.0) defines the strict conditions under which the theory would be empirically rejected, including quantitative thresholds for latency alignment, false structure detection, cross-linguistic stability, and external annotator agreement.

> *Important: The four semantic forces introduced in DSFT are operational modeling constructs, not claims about biological cognition, quantum mechanics, or consciousness. All reported performance figures are from controlled synthetic benchmarks; real-world validation is planned and ongoing.*

Keywords: *semantic field theory, temporal dialogue modeling, force dynamics, NLP, transition detection, observer effects, DSFT, IKPS-CORE, entropic linguistics*

1. Introduction

1.1 The Problem with Static Classification

Contemporary NLP systems are predominantly designed around the assumption that meaning is a property of individual utterances. A sentence is embedded, classified, or labeled, and the analysis is complete. This architecture is well-suited for tasks such as sentiment classification, topic detection, or named entity recognition, where the unit of analysis is indeed the sentence or document.

Yet human dialogue is fundamentally different. Meaning in conversation is not instantiated at the level of the turn — it emerges from the temporal dynamics between turns. A dialogue that begins as a rigorous technical exchange may gradually accumulate affective urgency; an exploratory philosophical discussion may drift toward persuasive rhetoric. These transitions are processes, not events, and they cannot be captured by systems that treat each utterance independently.

The inadequacy of static approaches manifests in at least four observable failure modes: (a) inability to detect gradual semantic drift, (b) no advance warning before dominance transitions, (c) loss of contextual continuity when long-form dialogue exceeds the model's effective attention window, and (d) artificial discreteness in outputs that obscure continuous underlying dynamics.

1.2 From Classification to Field Theory

DSFT proposes a fundamental reconceptualization: instead of asking 'what is this utterance?', we ask 'how are the forces within this dialogue evolving?'. The metaphor is drawn deliberately from classical field theory — a domain in which the behavior of any point in a field is determined not by an isolated local property but by the configuration of forces acting upon it from all directions.

In dialogue, we identify four such forces. Analytical Pressure captures the directional tendency toward logical structure, deductive reasoning, and factual precision. Exploratory Expansion captures the outward movement toward possibility, hypothesis, and open-ended inquiry. Affective Resonance captures the weight of emotional valence — concern, urgency, distress, or warmth — that colors the semantic content. Persuasive Drift captures the rhetorical pull of argumentation toward directed conclusions.

Crucially, these forces do not operate in isolation. They are coupled — analyticity suppresses exploration, exploration amplifies affect, affect resists persuasion, persuasion constrains analyticity. The result is a dynamic semantic field whose temporal evolution traces the semantic arc of the conversation.

1.3 Contributions of This Work

This paper makes the following contributions:

- A formal mathematical specification of the DSFT force evolution equation, incorporating inertia, inter-force coupling, momentum, and hysteresis resistance.
- A seven-stage developmental history documenting the iterative failure-and-refinement process by which DSFT was derived from its precursor systems (VEFS, SBCL, RFDM, RFDM-I, RFDM-II).

- Empirical results from controlled synthetic benchmarks demonstrating 7-turn early transition detection and 3.3% false alarm rate.
- A configurable observer layer with four distinct measurement modes that alter field sampling behavior.
- A complete external falsification protocol (EVALUATION_PROTOCOL v1.0) specifying conditions under which the theory is rejected.
- An open-source reference implementation (IKPS-CORE v2.1.0) with full benchmark suite and reproducibility configuration.

1.4 Positioning Relative to Existing Work

DSFT does not compete with transformer-based language models. Models such as BERT [Devlin et al., 2019] and GPT-family architectures excel at token-level and sentence-level semantic representation. They do not, however, provide an interpretable temporal layer for tracking how semantic emphasis evolves across a dialogue as a whole. DSFT is designed as a complementary layer that operates above token-level representations, providing a force-dynamic interpretation of discourse evolution.

The closest relatives in the literature are hidden Markov models (HMMs) applied to dialogue state tracking [Young et al., 2013], topic models applied to discourse segmentation [Blei et al., 2003], and computational approaches to sentiment dynamics [Pang & Lee, 2008]. DSFT differs from all of these in that it explicitly models inter-force coupling as a continuous differential system, tracks momentum and resistance as first-class temporal quantities, and provides an observer-configurable measurement layer.

2. Theoretical Framework

2.1 The Epistemic Projection Space (EPS)

DSFT is embedded within a broader architectural concept: the Integrated Knowledge Projection System (IKPS), which frames all knowledge representation as a projection rather than a storage operation. The foundational operator is the Epistemic Projection Operator:

$$\hat{P}: D(t) \rightarrow EPS(t)$$

Equation 1: The Epistemic Projection Operator maps dialogue state D(t) to the Epistemic Projection Space at time t.

The critical architectural constraint is the No-Storage Condition, which demands that no two temporal snapshots of the EPS overlap:

$$\forall\ t_1 \neq t_2 : EPS(t_1) \cap EPS(t_2) = \emptyset$$

Equation 2: The No-Storage Condition ensures each projection is temporally distinct.

This condition distinguishes between archival persistence (prohibited) and transient computational state (permitted). The EPS at any given moment encodes the current configuration of the semantic force field, not a cumulative history of prior states. This design choice enforces a fundamentally streaming, non-archival epistemology.

The geometric structure of EPS remains an open research question. Candidate structures include a Riemannian manifold with curvature determined by force coupling, a Hilbert-like functional space with force values as basis coefficients, or a probabilistic latent space in the sense of variational representation. Future theoretical work will formalize this geometry.

2.2 The Four Semantic Forces

DSFT defines four operational semantic forces. The following table provides complete specifications:

Force	Symbol	Semantic Domain	Key Markers	Coupling Sign
Analytical Pressure	F_A	Logic, deduction, precision	jacobian, eigenvalues, therefore, implies	Negative to F_E, F_R
Exploratory Expansion	F_E	Possibility, hypothesis, inquiry	perhaps, maybe, explore, what if	Positive to F_R
Affective Resonance	F_R	Emotion, urgency, concern, care	concern, danger, crisis, urgent	Positive to F_P
Persuasive Drift	F_P	Rhetoric, directed conclusion	guaranteed, prove, customers, must	Positive to F_A

Table 1: Specification of the four DSFT semantic forces as operational modeling constructs.

> *These forces are operational constructs for modeling dialogue dynamics. They do not claim to represent underlying cognitive processes, neural states, or phenomenological categories.*

2.3 Core Hypothesis

The central claim of DSFT is that semantic meaning in dialogue is not a static property of individual utterances but the dynamic result of force interactions across time. Formally:

> *"Meaning is not a point in space. It is the dynamics of interaction between opposing forces."*

This hypothesis generates a testable prediction: if DSFT correctly models semantic dynamics, then the state of the force field at time t should contain predictive information about the dominant force at time t+k for some k > 0. The empirical claim of the present work is that k = 7 turns in controlled synthetic benchmarks.

3. Mathematical Formulation

3.1 Force Evolution Equation

The temporal dynamics of each semantic force F_i are governed by the following discrete-time evolution equation:

$$F\_i(t+1) = \alpha \cdot F\_i(t) + \beta \cdot \Sigma_j\ C_{ij} \cdot F_j(t) + \gamma \cdot M_i(t) - \lambda \cdot R_i(t) + \varepsilon_i(t)$$

Equation 3: The DSFT force evolution equation. All four forces evolve simultaneously under this dynamics.

Each term in the equation contributes a distinct physical interpretation:

Term	Symbol	Value	Role	Interpretation
Inertia	$\alpha \cdot F\_i(t)$	$\alpha = 0.2$	Memory of past state	Forces tend to persist; resistance to rapid change
Coupling	$\beta \cdot \Sigma\ C_{ij} \cdot F_j(t)$	$\beta = 0.25$	Inter-force interaction	Each force amplifies or suppresses others via the coupling matrix
Momentum	$\gamma \cdot M_i(t)$	$\gamma = 0.5$	Rate-of-change amplification	Forces moving strongly in one direction tend to continue
Resistance	$\lambda \cdot R_i(t)$	$\lambda = 0.1$	Hysteresis penalty	Dominant forces accumulate resistance, enabling transitions
Stochastic	$\varepsilon_i(t)$	$\sim N(0,\sigma)$	Noise injection	Prevents degenerate steady states; models real dialogue variability

Table 2: Decomposition of the force evolution equation with calibrated parameter values.

3.2 Momentum Term

The momentum of force F_i at time t is defined as the finite difference:

$$M_i(t) = F_i(t) - F_i(t-1)$$

Equation 4: Semantic momentum captures the direction and rate of force change.

Momentum ensures that forces with rising trajectories will continue to rise unless actively opposed by coupling or resistance terms. This property is critical for early transition detection: a force may have a low absolute value while simultaneously having a high positive momentum, signaling an imminent dominance shift.

3.3 Hysteresis Resistance

Forces that have been dominant for extended periods accumulate resistance:

$$R_i(t) = \text{persistence\_count}(F_i) \times 0.15$$

Equation 5: Hysteresis resistance increases proportionally to the number of turns a force has been dominant.

This mechanism prevents entrenchment: a force that dominates for many turns will accumulate sufficient resistance to eventually yield to a successor force. The hysteresis coefficient (0.15) was calibrated empirically to produce realistic transition dynamics without inducing rapid oscillation.

3.4 Coupling Matrix

The inter-force coupling is specified by a 4×4 asymmetric matrix C:

```
C = [[0, -0.15, -0.10, 0.12],
     [-0.12, 0, 0.18, -0.08],
     [-0.08, 0.15, 0, 0.10],
     [0.10, -0.05, 0.08, 0]]
```

Equation 6: The DSFT coupling matrix C. Rows and columns correspond to [F_A, F_E, F_R, F_P].

Key coupling relationships encoded in this matrix:

- F_A → F_E: C[0,1] = -0.15 — Analytical pressure suppresses exploration (reduction in uncertainty-seeking when logical structure is dominant).
- F_E → F_R: C[1,2] = +0.18 — Exploration amplifies affect (open inquiry invites emotional investment).
- F_R → F_A: C[2,0] = -0.08 — Affective resonance weakly suppresses analytical pressure.
- F_P → F_A: C[3,0] = +0.10 — Persuasion leverages analytical structure.

3.5 Precursor Probability and Transition Detection

Early transition detection is achieved through a precursor probability function that monitors each force for signs of imminent dominance:

$$P\_\text{precursor}(F_j) = \text{mean}(F_j\_\text{recent}) + \max(0, \text{trend}) \times 2 + \text{residual}_j$$

Equation 7: Precursor probability for force F_j. Detection fires when P > 0.5.

The three components of P_precursor capture different aspects of the approaching transition:

- mean(F_j_recent): the recent absolute level of the force — a high baseline indicates proximity to dominance.
- max(0, trend) × 2: the upward momentum component, amplified to ensure rising forces register early warning.
- residual_j: the accumulated residual from forces that have receded, which can transfer predictive weight to their successors.

3.6 Transition Latency

The transition latency L is defined as:

$$L = t\_dominance - t\_precursor$$

Equation 8: Transition latency measures how many turns before dominance the precursor signal was detected.

A positive value of L indicates early detection — the system identified the transition before it occurred. The empirical result from controlled benchmarks is L = 7 turns across all five tested transition types. A negative value would indicate late detection, which constitutes a failure mode under the external evaluation protocol.

4. System Evolution: Seven Stages of Development

DSFT was not conceived in a single design step. It emerged through seven distinct developmental stages, each motivated by the failure of its predecessor. This iterative failure-driven methodology is documented here both for scientific transparency and as a contribution to understanding how complex dynamic systems can be progressively refined.

Stage	System	Key Innovation	Result	Failure / Achievement
1	VEFS	Vector space + orthogonal basins	2/4 correct	Basin overlap between emotional and exploratory
2	SBCL	Semantic basis operators	1/4 correct	Complete collapse to single dominant force
3	RFDM	Relational force dynamics	4/4 correct	First full differentiation — no temporal continuity
4	RFDM-I	Inertia added to RFDM	4/4 + stability	Stability measurable only as zero (unmeasurable)
5	RFDM-II	Coupling + stochastic noise	4/4 + σ=0.5	Measurable stability, correlations detected
6	Observer Layer	Configurable measurement modes	4 observer modes	Observer effect quantified
7	DSFT-TD	Momentum + residual tracking	L=7 early detection	Full temporal dynamics — current stable core

Table 3: Seven-stage developmental history of DSFT, documenting the failure-driven refinement process.

4.1 Stage 1: VEFS — Vectorized Epistemic Field System

VEFS represented semantic types as orthogonal basins in a vector field. The system successfully classified Technical (Analytical) and Exploratory dialogue types but confused Emotional and Exploratory categories due to basin overlap. This failure revealed that orthogonality constraints alone are insufficient to separate semantic dimensions that share surface-level markers.

4.2 Stage 2: SBCL — Semantic Basis Construction Layer

SBCL replaced basins with algebraic basis operators. The system exhibited catastrophic collapse: all four dialogue types were classified as Technical/Analytical, revealing that the operator algebra was insufficiently constrained to prevent degenerate fixed-point solutions. Only 1 of 4 dialogue types was correctly classified.

4.3 Stage 3: RFDM — Relational Field Dynamics Model

RFDM introduced the key conceptual innovation: forces as relational quantities rather than independent coordinates. For the first time, all four dialogue types were correctly classified (4/4). However, the system had no temporal continuity — each utterance was processed as an isolated snapshot with no memory of prior states.

4.4 Stage 4: RFDM-I — Inertia Addition

RFDM-I added an inertia term (α) to the force evolution equation, providing memory of prior states. Classification remained perfect (4/4) and forces demonstrated persistence across turns. However, the stability metric collapsed to zero, making the system's internal coherence unmeasurable — a technically valid but diagnostically opaque system.

4.5 Stage 5: RFDM-II — Coupling and Stochastic Noise

RFDM-II introduced the coupling matrix C and stochastic noise term ε. The coupling encoded inter-force relationships for the first time; noise injection prevented degenerate fixed-point attractors. The result was measurable stability ($\sigma = 0.5$ per force) and detectable inter-force correlations — a richer and more realistic dynamics.

4.6 Stage 6: Observer Layer

The observer layer addressed a philosophically and empirically important question: does the mode of measurement affect the field being measured? Four observer configurations were defined (Passive, Active, Reflexive, Meta) that alter the weighting and amplification of force values during sampling. This is not a claim about quantum measurement but a configurable architectural feature for researcher-controlled sensitivity analysis.

4.7 Stage 7: DSFT-TD — Transition Dynamics

The final stage introduced momentum tracking (M_i), residual signals, and the precursor probability function $P_{precursor}$. The result was the first system capable of detecting semantic transitions before they occurred in the dominant force. The controlled benchmark result — $L = 7$ turns early detection — represents the principal empirical contribution of this work.

5. System Architecture

5.1 Processing Pipeline

The DSFT-TD V2.1 pipeline consists of four sequential layers:

Layer	Component	Input	Output	Key Module
1	Marker Detection	Raw utterance text	Force intensity scores	forceDisentanglement.js
2	Force Dynamics Engine	Intensity scores + history	Evolved force values F_i(t+1)	dsft_td_v2.js
3	Precursor Detection	Force values + momentum	Early warning signals	earlyPredictor.js
4	Observer Layer	Force field state	Observer-modulated output	transitionMatrix.js

Table 4: DSFT-TD V2.1 processing pipeline with module assignments.

5.2 Marker Detection Layer

The marker detection layer extracts lexical signals associated with each semantic force from raw utterance text. The current implementation uses a curated lexical inventory — for example, words such as 'jacobian', 'eigenvalues', and 'therefore' activate Analytical Pressure; words such as 'concern', 'danger', and 'crisis' activate Affective Resonance. Force intensity is computed as a normalized count of activated markers per turn.

The marker-based approach represents the current first approximation. As documented in the limitations and future work sections, marker-free contextual detection — using semantic embeddings rather than surface-level cues — is the primary target for the next development phase.

5.3 Observer Layer Configuration

The observer layer offers four distinct measurement modes:

Observer Mode	Mechanism	Effect on Field	Measured Deviation	Use Case
PASSIVE	No modification to force values	Neutral measurement	0.0000	Baseline evaluation
ACTIVE	Amplifies the currently dominant force	Strengthens dominant signal	0.0669	Emphasis analysis

REFLEXIVE	Boosts weak/suppressed forces	Elevates minority forces	0.0000	Sensitivity to emerging forces
META	Recursive: applies observer to its own output	Self-referential amplification	0.0199	Observer effect studies

Table 5: Observer layer modes with quantified deviations from the passive baseline.

5.4 Repository Structure

The IKPS-CORE repository is organized as follows:

```
IKPS-CORE/
├── src/transition/
│   ├── dsft_td_v2.js          # Core dynamics engine
│   ├── semanticMomentum.js    # Momentum tracking
│   ├── transitionEntropy.js   # Turbulence measurement
│   ├── hysteresis.js          # Resistance system
│   └── earlyPredictor.js      # Precursor detection
├── benchmarks/                # Full validation suite
├── baselines/                 # Keyword + pattern baselines
├── validation/                # Real-data validator
└── config/benchmark.config.js # Centralized config (seed=42)
```

6. Experimental Results

> *All results in this section were obtained from controlled synthetic benchmarks under fixed configuration (seed=42, deterministic mode=true). Real-world validation on human-annotated corpora is planned as described in Section 8.*

6.1 Transition Detection Performance

DSFT-TD V2.1 was evaluated on five canonical semantic transition types, each modeled as a 20-turn synthetic dialogue in which the dominant force shifts from one category to another:

Transition Type	Precursor Detected	Dominance Shift	Latency (turns)	Status
Analytical → Affective	Turn 0	Turn 7	7	Early Detection ✓
Analytical → Persuasive	Turn 0	Turn 7	7	Early Detection ✓
Affective → Persuasive	Turn 0	Turn 7	7	Early Detection ✓
Persuasive → Exploratory	Turn 0	Turn 7	7	Early Detection ✓
Exploratory → Analytical	Turn 0	Turn 7	7	Early Detection ✓
Average	—	—	7.0	All Transitions Detected

Table 6: Transition detection performance across five canonical transition types (controlled benchmark).

6.2 Precursor Probability Progression

The following table illustrates how the precursor probability P(Affective) evolves across turns in an Analytical → Affective transition. The precursor signal crosses the detection threshold ($P > 0.5$) at turn 12, seven turns before the Affective force achieves dominance at turn 19:

Turn	P(Affective)	Dominant Force	Signal State
6	0.05	Analytical	Quiet — no signal
8	0.05	Analytical	Quiet — stable analytical
10	0.18	Analytical	Early precursor signal
12	0.38	Analytical	Emerging — below threshold
14	0.90	Analytical	Strong precursor — threshold exceeded
16	0.11	Affective	Post-transition — force established

Table 7: Precursor probability P(Affective) across turns in a canonical Analytical → Affective transition.

6.3 False Alarm Rate

False alarm resistance was evaluated by applying DSFT-TD to a 30-turn stable technical dialogue in which the dominant force remained Analytical throughout. The system reported Analytical dominance in 29 of 30 turns, producing a single false transition event:

Test Configuration	Duration	True Dominant	False Alarms	False Alarm Rate
Stable Technical Dialogue	30 turns	Analytical (29/30)	1 transition	3.3%

Table 8: False alarm rate on stable technical dialogue (controlled conditions).

6.4 Long-Form Stability

The system was evaluated across four extended dialogue scenarios to verify stability over long-form conversation:

Test Scenario	Duration	Result	Key Observation
Stable Technical	20 turns	Stable	90% Analytical dominance, 4 minor transitions
Gradual Transition	20 turns	Smooth transition	Continuous force evolution without discontinuity
Chaotic Oscillation	30 turns	No collapse	86.2% change rate — oscillation without degeneration
Semantic Drift	40 turns	Stable	1 major transition, sustained coherence

Table 9: Long-form stability results across four extended dialogue scenarios.

6.5 Comparison with Baselines

DSFT-TD V2.1 was compared against two non-temporal baselines under identical controlled conditions:

System	Accuracy (controlled)	Early Detection	False Alarm Rate	Temporal Model
Keyword Baseline	83.3%	None	N/A	No
Pattern Baseline	83.3%	None	N/A	No
DSFT-TD V2.1	100%	7 turns prior	3.3%	Yes — full dynamics

Table 10: Comparison of DSFT-TD V2.1 against keyword and pattern baselines (controlled benchmark). Note: Baseline comparison with transformer models (BERT, RoBERTa, LSTM, HMM) is planned for future work.

6.6 Real-Data Preliminary Validation

A preliminary real-data validation was conducted on three naturally occurring dialogue samples. The key finding was consistency in the directionality of dynamics: DSFT produced non-random force trajectories on real dialogues, though with higher transition rates than synthetic benchmarks (1.7% average vs. 0.0% in synthetic stable dialogues). The sample size (n=3) is insufficient for statistical claims; this result motivates the full validation protocol described in Section 7.

7. External Falsification Protocol

> *"This protocol exists to falsify DSFT-TD, not validate it. Any positive result is not considered success, but rather: survivability under external conditions." — EVALUATION_PROTOCOL v1.0*

7.1 Protocol Philosophy

Scientific credibility requires that DSFT be exposed not to conditions that favor it but to conditions designed to break it. The External Falsification Protocol (EFP v1.0) defines a rigorous empirical evaluation framework under which DSFT is evaluated as a hypothesis about temporal semantic structure, with explicit falsification criteria.

7.2 Evaluation Metrics

The EFP defines four primary evaluation metrics:

Metric	Symbol	Formula	Falsification Threshold
Latent Alignment Delay	LAD	t_human_transition - t_dsft_detection	Consistently negative over 10+ dialogues
False Structure Detection Rate	FSDR	DSFT_transitions_not_in_human / Total_DSFT_transitions	FSDR > 30%
Cross-Linguistic Stability Index	CLSI	1 - Var(F_AR, F_FR, F_EN) / MaxVariance	CLSI < 0.6
External Agreement Rate	EAR	Agreements / Total_turns	EAR < 0.5 on human-annotated data

Table 11: External falsification metrics with rejection thresholds.

7.3 Falsification Criteria

DSFT-TD is formally considered falsified if any of the following conditions are satisfied:

- CLSI < 0.6 on any language pair (English, Arabic, French) — indicating that the force model is language-specific rather than semantic.
- FSDR > 30% on any dialogue type — indicating that DSFT is detecting illusory structure not present in human annotation.
- LAD consistently negative over 10 or more dialogues — indicating systematic late detection rather than early warning.
- EAR < 0.5 on human-annotated data — indicating performance at or below chance agreement with human semantic judgment.

- Complete force collapse to a single dominant force over 50+ turn conversations — indicating that the dynamics degenerate to the same failure mode as SBCL.

7.4 Dataset Requirements

The EFP specifies minimum dataset requirements for valid external evaluation:

Language	Minimum Turns	Minimum Dialogues	Annotators	Min. κ
English	5,000	100	≥ 3 blind	0.70
Arabic	2,000	40	≥ 3 blind	0.70
French	2,000	40	≥ 3 blind	0.70

Table 12: Minimum dataset requirements for external falsification evaluation.

7.5 Data Independence Principle

To ensure genuine falsifiability, evaluation data must satisfy:

$$\mathtt{DATA} \notin \mathtt{TRAINING\_DATA}$$
$$\mathtt{DATA} \notin \mathtt{GENERATED\_BY\_DSFT}$$
$$\mathtt{DATA} \notin \mathtt{ALIGNED\_WITH\_DSFT\_FORCES}$$

Equation 9: The Data Independence Conditions for external evaluation.

Permitted sources include Reddit/forum archives, debate transcripts, therapy dialogues (with ethical approval), technical discussion boards, and multilingual human discourse corpora. Prohibited sources include any synthetic dialogues generated by DSFT or designed with awareness of the four semantic force categories.

8. Limitations and Future Work

8.1 Current Limitations

The following limitations must be understood when interpreting DSFT-TD V2.1 results:

Limitation	Description	Impact	Mitigation Planned
Synthetic data	Primary validation on controlled, curated dialogues	Results may not generalize to natural discourse	EFP real-data evaluation
Sample size	Real-data pilot: n=3 samples only	Insufficient for statistical inference	Minimum n=100 dialogues planned
Marker dependence	Force detection uses lexical markers	Misses semantic signals without surface markers	Embedding-based marker-free detection
Language	English only (validation)	Force model may be language-specific	Arabic, French validation (EFP)
Length bound	Tested to 40 turns maximum	Behavior at 50+ turns untested	Extended benchmark suite
Baseline limited	Compared to keyword/pattern only	Cannot quantify advantage over neural models	BERT, RoBERTa, LSTM comparison planned
EPS geometry	Geometric structure of EPS undefined	Theoretical foundations partially informal	Manifold / Hilbert structure formalization

Table 13: Current limitations of DSFT-TD V2.1 with planned mitigations.

8.2 Immediate Research Priorities (0–3 months)

- Real-data validation on Reddit debate archives and structured argumentation corpora using the EFP pipeline.
- Baseline expansion: implementation of BERT, RoBERTa, LSTM, and HMM comparison under identical controlled conditions.
- Reproducibility pack release (v2.1.1): fixed seeds, deterministic configuration files, and exact dataset snapshots for complete replication.
- Public evaluation suite: open benchmark release enabling independent replication by the research community.

8.3 Medium-Term Research Directions (3–12 months)

- Marker-free emergence: replacing lexical marker detection with contextual semantic embedding detection, enabling force recognition without surface-level cues.
- Multilingual validation: Arabic and French force dynamics using parallel corpora and native-speaker annotation.
- Adversarial robustness: testing DSFT against dialogues deliberately designed to confuse force classification through contradiction, paradox, and mixed-register text.
- EPS geometric formalization: developing the mathematical structure of the Epistemic Projection Space as a manifold, Hilbert space, or probabilistic latent space.
- Interactive visualization dashboard: force field trajectories rendered as real-time temporal plots for qualitative analysis and demonstration.

9. Reproducibility

9.1 Fixed Configuration

All benchmarks in this paper were run under the following fixed configuration:

Parameter	Value
Seed	42
Deterministic mode	true
α (inertia)	0.2
β (coupling)	0.25
γ (momentum)	0.5
λ (resistance)	0.1
residualDecay	0.7
transitionThreshold	0.35

Table 14: Fixed configuration for all reported benchmark results.

9.2 Benchmark Execution

All benchmarks can be reproduced with the following commands:

```
git clone https://github.com/gitdeeper12/IKPS-CORE.git
cd IKPS-CORE && npm install
npm run benchmark:all            # Full suite
npm run benchmark:transitions    # Transition detection
npm run benchmark:latency        # Latency measurement
npm run benchmark:drift          # Drift prediction
npm run benchmark:stability      # Long-form stability
npm run validate:real            # Real-data validation
npm run test:reproducibility     # Verify deterministic output
```

9.3 Open Science Distribution

IKPS-CORE is distributed across the following open science platforms:

Platform	Location	Content
GitHub	https://github.com/gitdeeper12/IKPS-CORE	Source code, benchmarks, baselines
GitLab	https://gitlab.com/gitdeeper12/IKPS-CORE	Mirror repository
Zenodo	https://doi.org/10.5281/zenodo.20303214	Archived release with DOI
License	MIT License	Full open-source distribution

Table 15: Open science distribution platforms for IKPS-CORE.

10. Connections to the EntropyLab Research Program

DSFT emerges from and contributes to the broader EntropyLab decadal research program, which applies thermodynamic and information-theoretic entropy frameworks to AI systems, physics, and control theory. Several conceptual connections to prior EntropyLab work are noted:

10.1 Semantic Entropy and Force Transitions

The transition dynamics in DSFT share formal structure with the entropic phase transition models developed in ENTRO-AI (E-LAB-02), which modeled LLM hallucination as thermodynamic phase transitions. In DSFT, dominant force changes can be interpreted as first-order phase transitions in the semantic field, with the precursor probability P_precursor functioning as an order parameter that signals proximity to the critical point.

10.2 Observability Index and Observer Layer

The DSFT observer layer conceptually parallels the Observability Index Ω introduced in Irreducible Path Entropy in Neural Networks (E-LAB-11 adjacent). Both constructs formalize the degree to which an external measurement process perturbs the system being observed. In DSFT, the Active observer mode (deviation 0.0669) provides an empirical estimate of observation-induced perturbation.

10.3 Coupling and the Grand Constraint Network

The DSFT coupling matrix C is structurally analogous to the constraint connectivity encoded in the Grand Constraint Network (GCN) of NEUROPIA (E-LAB-10). Both systems represent interactions between functional subsystems as asymmetric coupling coefficients; both produce emergent macroscopic behaviors (semantic dominance in DSFT, physical-domain unification in NEUROPIA) that are not reducible to the properties of any single subsystem.

11. Conclusion

Dynamic Semantic Field Theory (DSFT) represents a principled departure from the dominant paradigm of static, utterance-level text classification. By reframing dialogue semantics as a continuously evolving field of four interacting forces — Analytical Pressure, Exploratory Expansion, Affective Resonance, and Persuasive Drift — DSFT provides the theoretical machinery to detect, characterize, and predict semantic transitions before they occur.

The DSFT-TD V2.1 implementation achieves the following results under controlled benchmark conditions: correct four-force dominance classification across all synthetic dialogue types, early transition detection at L = 7 turns before dominance shift, a false alarm rate of 3.3% in stable dialogues, and sustained stability over 40+ turn conversations. A configurable observer layer allows researcher-controlled sensitivity analysis across four measurement modes.

Equally important to these empirical results is the methodological commitment embodied in the External Falsification Protocol v1.0: DSFT is presented not as a validated theory but as a hypothesis that has survived controlled synthetic evaluation and must now survive external falsification on human-annotated natural language corpora. The protocol's quantitative rejection criteria — FSDR > 30%, CLSI < 0.6, EAR < 0.5 — provide the scientific community with clear conditions under which DSFT should be abandoned.

The IKPS-CORE open-source repository provides a fully reproducible implementation with fixed seeds, complete benchmark suite, baseline comparisons, and real-data validation infrastructure. Future work will expand this foundation through transformer-scale baseline comparisons, multilingual validation, marker-free contextual detection, and geometric formalization of the Epistemic Projection Space.

> *The semantic forces are operational modeling constructs for analyzing dialogue dynamics, not claims about biological cognition, quantum measurement, or consciousness. All performance figures are from controlled synthetic benchmarks; real-world validation is planned per the External Falsification Protocol v1.0.*

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Appendix A: Complete Parameter Specification

Parameter	Symbol	Value	Role	Source
Inertia coefficient	α	0.2	Memory of past force state	Calibrated RFDM-I
Coupling strength	β	0.25	Inter-force interaction weight	Calibrated RFDM-II
Momentum coefficient	γ	0.5	Rate-of-change amplification	Calibrated DSFT-TD
Resistance coefficient	λ	0.1	Hysteresis resistance strength	Calibrated DSFT-TD
Residual decay	—	0.7	Rate of residual signal decay	Calibrated DSFT-TD
Transition threshold	—	0.35	P_precursor threshold for early warning	Calibrated DSFT-TD
Random seed	—	42	Deterministic benchmark execution	Fixed

Table A1: Complete parameter specification for DSFT-TD V2.1.

Appendix B: Coupling Matrix (Full Specification)

The full 4×4 asymmetric coupling matrix C with row/column labeling [F_A, F_E, F_R, F_P]:

C_ij	→ F_A	→ F_E	→ F_R	→ F_P
F_A →	0	-0.15	-0.10	+0.12
F_E →	-0.12	0	+0.18	-0.08
F_R →	-0.08	+0.15	0	+0.10
F_P →	+0.10	-0.05	+0.08	0

Table B1: DSFT coupling matrix C. Entry C[i,j] is the effect of force F_i on force F_j per time step. Negative values indicate suppression; positive values indicate amplification.

Appendix C: Citation

BibTeX citation for this work:

```
@software{baladi2026dsft,
  author  = {Baladi, Samir},
  title   = {DSFT-TD V2.1: Dynamic Semantic Field Theory},
  year    = {2026},
  version = {2.1.0},
  doi     = {10.5281/zenodo.20303214},
  url     = {https://github.com/gitdeeper12/IKPS-CORE},
  license = {MIT}
}
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